

Paulson Alex Fernandes

azurepaulson06@gmail.com | +91 8459150386
<https://github.com/ScriptSynapse1> | <https://www.linkedin.com/in/paulson-fernandes-754710378>

Skills

Languages: Python, C, Java, SQL

Technologies & Tools: Wireshark, Kali Linux, Burp Suite, SQL, Git, HTML/CSS, Linux Terminal, Networking,

Soft Skills: Problem Solving, Communication, Team Collaboration

Work Experience

Secure Password Generator (GitHub Project)

Aug 2025 – September 2025

Project Developer

- Designed and implemented a password manager using HTML, CSS, and JavaScript with built-in cryptography for secure storage.
- Enhanced password safety by enabling strong password generation and encrypted saving of credentials.
- HTML, CSS, JavaScript, Cryptography, and GitHub.

Cybersecurity (Certification – Launched Global)

Aug 2025 – September 2025

Learner

- Built a solid foundation in networking and security concepts, including OSI, TCP/IP, firewalls, and intrusion detection/prevention systems (IDS/IPS).
- Studied major threats and attack types such as malware, DoS/DDoS, MITM, and brute force.
- Learned security frameworks and methodologies, including the CIA Triad, Cyber Kill Chain, and risk/vulnerability assessments.
- Gained hands-on exposure to tools and standards such as Nmap, OpenVAS, SIEM, AES, RSA, hashing, OWASP Top 10, ISO 27001, NIST, and MITRE ATT&CK.
- Networking & Security, Threat Analysis, Risk Assessment, Nmap, OpenVAS, SIEM, Cryptography, Compliance Standards

Cybersecurity Intern at BTech Walleh

August 2025 – August 2025

- Investigated a Conti ransomware incident with Splunk, uncovering persistence, credential theft, and exploited CVEs, and recommended remediation.
- Performed penetration testing on Windows using Metasploit (Eternal Blue), achieving SYSTEM access and cracking password hashes.
- Analysed a Lumma Stealer campaign with Wireshark & Zeek, finding malicious domains, C2 activity, and exfiltration IOCs for SIEM integration.
- Simulated ARP poisoning and configured Windows Firewall rules to prove risks of unencrypted traffic and enforce endpoint security.
- Splunk, Wireshark, Zeek, Metasploit, Nmap, John the Ripper, Cain & Abel, Ettercap, Windows Firewall

Education

MIT-World Peace University

July

2025-Present

B.Tech. in Computer Science and Engineering (Cybersecurity and Forensics) ^[06]

Relevant Coursework: Introduction to Cybersecurity, Programming in C and Python, Data Structures and Algorithms, Computer Networks, Operating Systems, Digital Forensics, Cryptography

Project Work

- **Secure Password Generator (2025)**: Built a modern web tool that generates cryptographically strong passwords in the browser. Used the Web Crypto API's `crypto.getRandomValues` for randomness. Added features for adjustable password length (4-64 chars) and toggles for uppercase, lowercase, numbers, and symbols. Included a live strength meter based on entropy (length × character-set size). Implemented copy-to-clipboard functionality. All client-side; no server interaction. Stack: HTML, CSS, and vanilla JavaScript.
- **Conti Ransomware Investigation (Splunk)** – Investigated Conti ransomware attack using Splunk logs; found persistence mechanisms, credential theft (LSASS dumping), and web shell deployment. Documented IOCs and exploited CVEs (CVE-2018-13374/13379, CVE-2020-0796).
- **TryHackMe Blue (Eternal Blue Exploit)** – Executed reconnaissance and exploitation of MS17-010 (Eternal Blue) vulnerability; escalated privileges to SYSTEM, extracted password hashes, cracked credentials, and captured system flags.
- **Threat Analysis of Lumma Stealer (Wireshark & Zeek)** – Analysed PCAP traffic of fake Google Authenticator & Teams installers; found malicious ZIP payloads, C2 HTTP POST exfiltration, and extracted IOCs for detection and mitigation.
- **ARP Poisoning & File Capture (Wireshark)** – Performed ARP poisoning using Cain & Abel / Ettercap; intercepted HTTP file transfers and reconstructed captured data, proving the risk of unencrypted communications.
- **Windows Firewall Access Control** – Configured outbound rules in Windows Defender Firewall to block YouTube IPs; validated network restrictions using ping, nslookup, and browser tests.
- **AI-Based Insider Threat Detection System** – Developed an AI-powered insider threat detection platform using Isolation Forest, Random Forest, and XGBoost to analyze enterprise user behavior, detect anomalous activities, and generate real-time threat scores through a Flask-based monitoring dashboard and behavioral analytics pipeline.
- **SSH Honeypot Threat Intelligence Platform** – Built a full-stack Python honeypot (Paramiko) simulating a real Ubuntu server on a public VPS; captured 155,000+ real brute-force attempts across 14 days, logging credentials, post-auth commands, and malware downloads into SQLite; integrated 6 live threat intelligence feeds (Feodo Tracker, BlocklistDE, Emerging Threats, Spamhaus) with automatic GeoIP enrichment and cross-referencing; built a real-time WebSocket dashboard with world map, attack heatmap, PDF reporting, and JWT-authenticated admin panel.

Awards and Certificates

- **Launched Cybersecurity Certificate:**
- **Core Knowledge:** Networking & Security (OSI, TCP/IP, Firewalls, IDS/IPS), Threats & Attacks (Malware, DoS/DDoS, MITM, Brute Force), Security Concepts (CIA Triad, Cyber Kill Chain, Risk & Vulnerability Assessment)
- **Tools & Compliance:** Nmap, OpenVAS, SIEM, Foot printing, Cryptography (AES, RSA, Hashing, Email Security), Frameworks & Standards (OWASP Top 10, ISO 27001, NIST, MITRE ATT&CK)